

Hilbert-valued weighted Sobolev approximation

The missing uniform-tail condition and an exact primitive criterion

Autonomous solve-first investigation, lane 12

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Abstract

Pérez and Quintana ask for the closure of Hilbert-valued polynomials in a weighted $W^{1,\infty}$ space. We first repair the infinite-dimensional L^∞ input: coordinatewise scalar approximability is necessary but not sufficient; the missing condition is uniform decay of the weighted coordinate tails. This gives an exact closure theorem and a counterexample to the coordinatewise-only assertion in the companion paper. We then prove that whenever the weighted primitive operator is bounded, the Sobolev closure consists exactly of the functions whose derivative satisfies that corrected L^∞ criterion. A one-sided coordinate comparison is an explicit sufficient condition. In particular, for every constant positive diagonal Hilbert weight w , the answer is exactly $\{f : D_w f \in C^1(I; \mathcal{H})\}$. The unrestricted formulation also permits weights for which the displayed quantity is only a seminorm, so no claim is made for every weakly measurable weight.

1 Corrected setting and the weighted sup-norm closure

Let $I = [a, b]$, let $\mathcal{H}$ be a separable real Hilbert space with fixed orthonormal basis $(e_j)_{j \geq 1}$, and write $f_j = \langle f, e_j \rangle$. Let $w = (w_j)_{j \geq 1} \in L^\infty(I; \mathcal{H})$, with $w_j > 0$ almost everywhere. Multiplication is coordinatewise, as in the source, and

$$\|f\|_w = \operatorname{ess\,sup}_{t \in I} \left(\sum_{j \geq 1} |f_j(t) w_j(t)|^2 \right)^{1/2}.$$

Denote the resulting space by X_w . For each j , let

$$\mathcal{A}_j = \overline{C(I)}^{L^\infty(I, w_j)};$$

the scalar weighted Weierstrass theorem quoted in [1] describes this space by one-sided regularity and weighted matching at the singularities of w_j . Define

$$\mathcal{A}_w = \left\{ f \in X_w : f_j \in \mathcal{A}_j \ (j \geq 1), \quad \lim_{N \rightarrow \infty} \operatorname{ess\,sup}_t \sum_{j > N} |f_j(t) w_j(t)|^2 = 0 \right\}. \quad (1)$$

Theorem 1 (uniform-tail correction). *The closures in X_w of $C(I; \mathcal{H}) \cap X_w$ and of the $\mathcal{H}$ -valued polynomials $\mathcal{P}(\mathcal{H})$ both equal $\mathcal{A}_w$. In finite dimension the tail condition is automatic.*

Proof. Suppose first that $g \in C(I; \mathcal{H})$. Coordinate extraction is contractive from X_w to $L^\infty(I, w_j)$. Moreover, compactness of $g(I)$ gives $\sup_t \|(I - P_N)g(t)\| \rightarrow 0$, where P_N is coordinate projection. Hence

$$\operatorname{ess\,sup}_t \sum_{j > N} |g_j(t) w_j(t)|^2 \leq \|w\|_{L^\infty(I; \mathcal{H})}^2 \sup_t \|(I - P_N)g(t)\|^2 \rightarrow 0.$$

Both properties pass to X_w limits, proving necessity.

Conversely, let $f \in \mathcal{A}_w$ and fix $\varepsilon > 0$. Choose N so the tail in (1) has square root below $\varepsilon/2$. For $1 \leq j \leq N$, choose $g_j \in C(I)$ such that

$$\|f_j - g_j\|_{L^\infty(I, w_j)} < \frac{\varepsilon}{2\sqrt{N}}.$$

Then $g = \sum_{j=1}^N g_j e_j$ is continuous and $\|f - g\|_w < \varepsilon$. Finally, vector-valued Weierstrass approximation gives $p \in \mathcal{P}(\mathcal{H})$ uniformly close to g , and

$$\|g - p\|_w \leq \|w\|_{L^\infty(I; \mathcal{H})} \|g - p\|_{L^\infty(I; \mathcal{H})}.$$

Thus polynomials have the same closure. $\square$

The extra condition in (1) cannot be discarded.

Proposition 2 (coordinatewise approximation is insufficient). *Let $\mathcal{H} = \ell^2$, $I = [0, 1]$, and $w_j(t) = 2^{-j}$. There is $f \in X_w$ such that every f_j is continuous, but*

$$\text{dist}_{X_w}(f, C(I; \ell^2)) \geq \frac{1}{2}.$$

Consequently, the coordinatewise-only infinite-dimensional closure assertion in [2] is false as written.

Proof. Put $t_j = 2^{-j}$ and choose continuous triangular bumps ϕ_j of height one, centered at t_j , with pairwise disjoint supports and with 0 outside every support. Define $f_j = 2^j \phi_j$. At each t at most one coordinate is nonzero, so $f(t) \in \ell^2$, while $\|f\|_w = 1$ and every f_j is continuous. If $g \in C(I; \ell^2)$ and $\|f - g\|_w < 1/2$, evaluation at t_j gives

$$|1 - 2^{-j} g_j(t_j)| < \frac{1}{2}, \quad |g_j(t_j)| > 2^{j-1}.$$

The essential-supremum inequality holds at the centers as well because both sides of each coordinate inequality are continuous. This contradicts boundedness of the continuous function g on I . Equivalently, the uniform weighted tail of f is always one. $\square$

2 Exact Sobolev lifting by the primitive operator

To remove the representative ambiguity in the source definition, set

$$W_w^{1,\infty} = \{f \in AC(I; \mathcal{H}) : f, f' \in X_w\}, \quad \|f\|_{W_w} = \|f\|_w + \|f'\|_w,$$

where AC means Bochner absolutely continuous. Assume that the primitive $Vg(t) = \int_a^t g(s) ds$ obeys

$$\|Vg\|_w \leq C_V \|g\|_w \tag{2}$$

whenever $g \in X_w \cap L^1(I; \mathcal{H})$.

Theorem 3 (weighted primitive criterion). *Under (2),*

$$\overline{\mathcal{P}(\mathcal{H})}^{W_w^{1,\infty}} = \{f \in W_w^{1,\infty} : f' \in \mathcal{A}_w\}. \tag{3}$$

Thus (1) gives an exact coordinate-and-tail characterization of the Sobolev closure for every weight satisfying (2).

Proof. If $p_n \rightarrow f$ in $W_w^{1,\infty}$, then the polynomials p'_n converge to f' in X_w , so Theorem 1 gives $f' \in \mathcal{A}_w$. Conversely, if $f' \in \mathcal{A}_w$, choose $q_n \in \mathcal{P}(\mathcal{H})$ with $\|q_n - f'\|_w \rightarrow 0$, and define the polynomial

$$p_n(t) = f(a) + \int_a^t q_n(s) ds.$$

Then $p'_n = q_n$, while absolute continuity and (2) give

$$\|f - p_n\|_w = \|V(f' - q_n)\|_w \leq C_V \|f' - q_n\|_w \rightarrow 0.$$

This proves convergence in the full Sobolev norm. $\square$

Here is a directly checkable weight hypothesis. If, for some $C \geq 1$,

$$w_j(t) \leq Cw_j(s) \quad (a \leq s \leq t \leq b, j \geq 1) \quad (4)$$

outside a common null set, then Minkowski's inequality gives

$$\|(Vg)(t)w(t)\|_{\ell^2} \leq \int_a^t \left(\sum_j |w_j(t)g_j(s)|^2 \right)^{1/2} ds \leq C(b-a)\|g\|_w.$$

Thus (2) holds. In particular it holds for every constant weight.

Corollary 4 (complete constant-weight answer). *Let $w(t) = \alpha = (\alpha_j) \in \ell^2$, where every $\alpha_j > 0$, and let $D_\alpha e_j = \alpha_j e_j$. Then*

$$\overline{\mathcal{P}(\mathcal{H})}^{W_w^{1,\infty}} = \{f \in W_w^{1,\infty} : D_\alpha f \in C^1(I; \mathcal{H})\}.$$

Proof. For constant w , Theorem 1 says precisely that $h \in \mathcal{A}_w$ iff $D_\alpha h$ has a continuous $\mathcal{H}$ -valued representative. Indeed, one direction follows by uniform limits; for the other, approximate the finitely many coefficients of a vector polynomial by vectors in the dense range of D_α . Since D_α is bounded and f is absolutely continuous, $(D_\alpha f)' = D_\alpha f'$ almost everywhere. Apply Theorem 3. $\square$

3 Scope and obstruction to the unrestricted question

The source permits arbitrary weakly measurable Hilbert-valued weights. In that generality its displayed quantity need not be a norm: for $\mathcal{H} = \mathbb{R}^2$ and $w = (1, 0)$, the nonzero constant $f = (0, 1)$ has $\|f\|_w = 0$. The same weighted class contains representatives whose second coordinate has no derivative, so Sobolev membership is not representative independent without an additional convention. Even after imposing nondegeneracy, (2) can fail when a coordinate weight grows too fast relative to its earlier values. Theorems 1 and 3 therefore solve a broad corrected class and isolate the two remaining issues—well-posedness and primitive boundedness—but do not claim the most-general classification requested in [1].

Eight focused upgrade routes were tested. Exact-title, author, journal, scalar weighted-Sobolev, and citation searches on 13 August 2026 found no later solution of the stated Hilbert-valued Sobolev problem and no correction of the missing uniform-tail condition. Novelty is plausible, not certified.

References

- [1] D. Pérez and Y. Quintana, *A survey on the Weierstrass approximation theorem*, Divulg. Mat. 16 (2008), 231–247; arXiv:math/0611038.
- [2] Y. Quintana, *On Hilbert extensions of Weierstrass' theorem with weights*, J. Funct. Spaces Appl. 8 (2010), 201–213; arXiv:math/0611034.